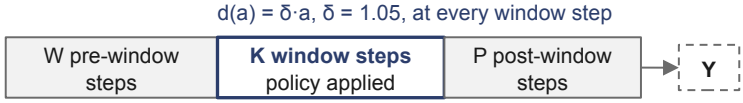
(A) Data and analysis sets

Representation set: 278 cases
Centres 2–5: 49 / 41 / 71 / 117
E1, E3-site, PCA and LEACE fitting

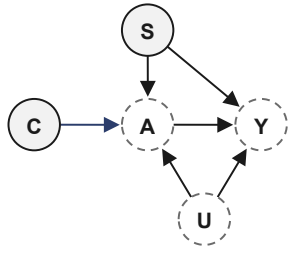
- **Primary causal set (W = 30): 158 cases**
  28 / 34 / 44 / 52
  - **Single-file subset: 123 cases**
    13 / 34 / 44 / 32 · sensitivity
- **W = 5 set: 201 cases**
  37 / 40 / 56 / 68 · sensitivity
- **W = 0 set: 204 cases**
  37 / 40 / 57 / 70 · sensitivity

(B) Semi-synthetic design (one case, one replicate)



1 step = 5 s of video; W and P steps follow their natural course

State S switches 0 → 1 at step s; window start
w = s + O, site-stratified offset O: −15, …, 15
Grid: K: 1, 2, 5, 15 · P: 0, 5, 30 · W: 0, 5, 30
α (unmeasured confounding): 0, 0.5, 1
γ (site loading): 0.5, 1
216 cells (+ 24 with autoregressive actions)
200 synthetic replicates per cell; site
loading c(C) permuted per replicate



solid = data; dashed = synthetic

(C) Adjustment arms

Arm	C	S	U	R	R, site removed
NV					
SITE	•				
ORACLE−C		•			
ORACLE	•	•			
FULL	•	•	•		
FM-P · FM-S · REF				•	
FM-P−C · FM-S−C · REF−C					•
FM-P-nat				•	

All arms also adjust for past synthetic actions H.

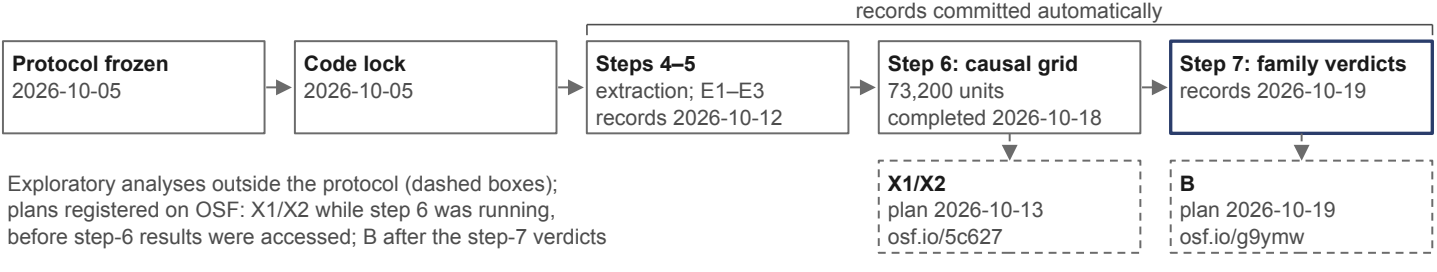
(D) Estimand, estimator and representations

Estimand. Ψ(δ) − Ψ(1) for the multiplicative modified treatment policy d(a) = δ·a (δ = 1.05) at each of the K window steps; true value exact for each replicate.

Estimator. Sequentially doubly robust; nuisances cross-fitted over 5 case-level folds; 95% Wald intervals from the influence function.

Representations. R = 64 principal components per step from SurgeNetCholec (FM-P), LeanVAE-4ch (FM-S) or Cosmos-1.0-Tokenizer-CV8x8x8 (REF); site removed by linear LEACE fitted on the 278-case training folds; FM-P-nat uses native frames.

(E) Workflow and registrations



Exploratory analyses outside the protocol (dashed boxes);
plans registered on OSF: X1/X2 while step 6 was running,
before step-6 results were accessed; B after the step-7 verdicts